DNS Troubleshooting and Network Diagnostic Tools

Course: The Bits and Bytes of Computer Networking
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1. Introduction

This lab aims to implement and analyze a basic network using network troubleshooting and diagnostic tools.

During the practice, several fundamental technologies and commands commonly used in IT environments and technical support were explored, including:

- DNS
- DHCP
- ping
- nslookup
- tracert
- name resolution
- connectivity testing

The lab simulates a simple corporate environment where a client PC must successfully communicate with a server using both IP addresses and domain names.

Troubleshooting tests were also performed by simulating DNS configuration failures in order to understand how they affect network connectivity.

2. Network Topology

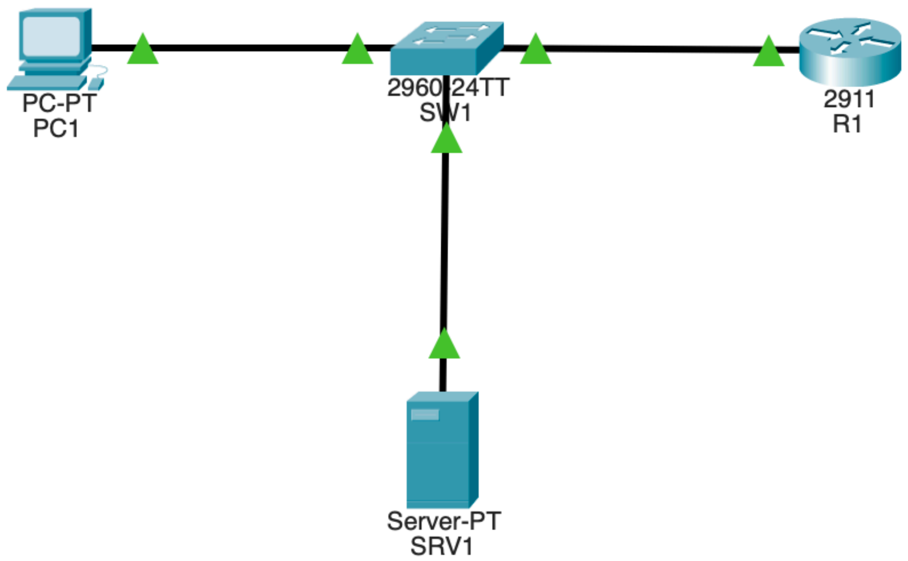
The network used in this lab consists of:

- One client PC
- One switch (SW1)
- One router (R1)
- One DNS/Web server

The connection is established as follows:

PC → Switch → Router → Server

Figure 1 – Network topology



3. Network Configuration

3.1 Router Configuration

The router was configured as the main gateway of the network using the following address:

- **IP Address:** 192.168.10.1
- **Subnet Mask:** 255.255.255.0

The interface is operating in an "up/up" state.

Figure 2 – Router IP configuration

```
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

3.2 DHCP Configuration

A DHCP pool was configured on the router to automatically assign IP addresses to network devices.

```
ip dhcp excluded-address 192.168.10.1
ip dhcp pool LAN
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 192.168.10.10
```

This configuration automatically assigns:

- IP address
- Subnet mask
- Default gateway
- DNS server

Figure 3 – DHCP configuration on the router

```
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.10.10
```

3.3 Server Configuration

The server was configured with:

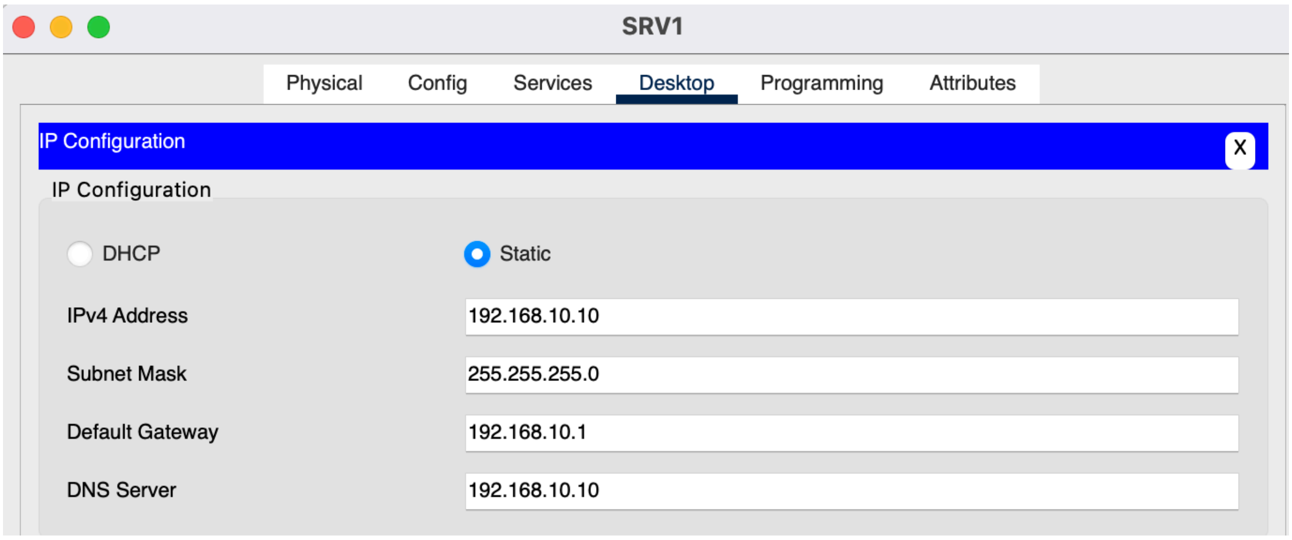
- Static IP address: 192.168.10.10
- DNS service enabled
- HTTP service enabled

Additionally, the following DNS record was created:

Domain Name	IP Address
empresa.com	192.168.10.10

This allows the domain name empresa.com to resolve to the server's IP address.

Figure 4 – Server DNS configuration



3.4 Client PC Configuration

The PC was configured in automatic DHCP mode in order to dynamically receive its network configuration from the router.

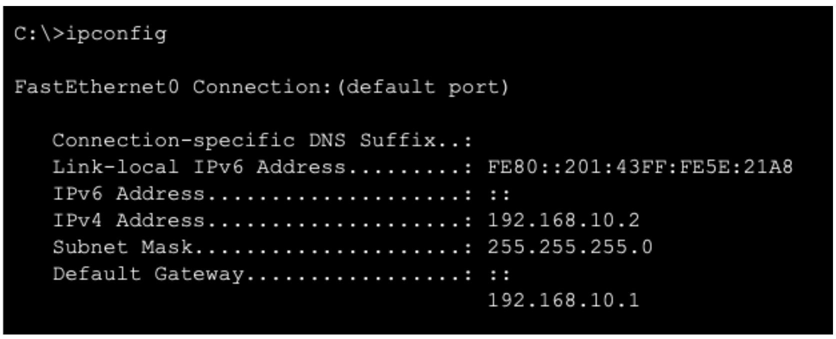
The configuration was verified using the following command:

ipconfig

Result obtained:

- IPv4 Address: 192.168.10.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Figure 5 – PC IP configuration



4. Connectivity and Troubleshooting Tests

4.1 Connectivity Test Using ping

Connectivity between the PC and the server was verified using:

ping 192.168.10.10

Result:

- Successful communication
- ICMP replies received correctly

Figure 6 – Successful ping to the server

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

4.2 DNS Resolution Using nslookup

The nslookup tool was used to verify DNS name resolution.

nslookup empresa.com

Result:

- The domain empresa.com was successfully resolved to 192.168.10.10

Figure 7 – Successful DNS resolution

```
C:\>nslookup empresa.com

Server: [192.168.10.10]
Address:  192.168.10.10

Non-authoritative answer:
Name:   empresa.com
Address:   192.168.10.10
```

4.3 Connectivity Test Using Domain Name

After validating DNS functionality, connectivity testing was performed using the domain name.

ping empresa.com

Result:

- Successful name resolution
- Successful communication with the server

Figure 8 – Ping using DNS name

```
C:\>ping empresa.com

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4.4 DNS Failure Simulation

To simulate a name resolution problem, an incorrect DNS server was manually configured on the client PC.

The following command was executed again:

nslookup empresa.com

Result:

- DNS resolution failure
- DNS server timeout

However, connectivity using the IP address continued to work:

ping 192.168.10.10

This demonstrates the difference between:

- Network connectivity problems
- DNS resolution problems

Figure 9 – DNS resolution failure

```
C:\>nslookup empresa.com

Server: [1.1.1.1]
Address:  1.1.1.1
DNS request timed out.
                 timeout was 15000 milli seconds.
DNS request timed out.
                 timeout was 15000 milli seconds.
DNS request timed out.
                 timeout was 15000 milli seconds.
```

4.5 DNS Problem Correction

The correct DNS configuration was restored and functionality was verified again.

The following tests were performed:

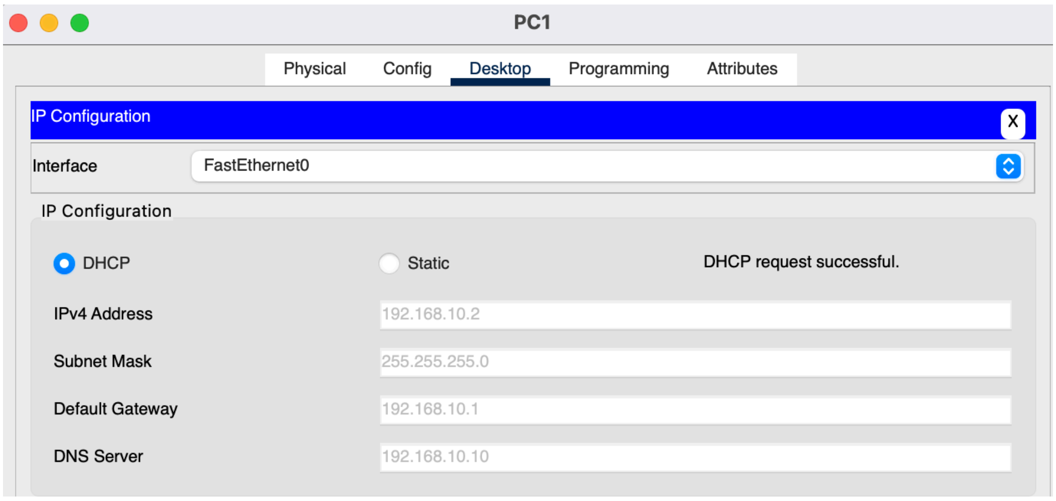
nslookup empresa.com

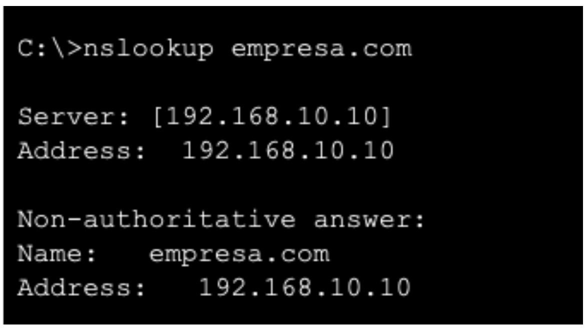
ping empresa.com

Result:

- Successful DNS resolution
- Connectivity restored

Figure 10 – DNS correction and restored connectivity





4.6 Route Tracing with tracert

The following command was used:

tracert empresa.com

Result obtained:

Tracing route to 192.168.10.10 over a maximum of 30 hops:

1   0 ms   0 ms   0 ms   192.168.10.10

Trace complete.

The tracert command allows visualization of the route followed by packets using TTL field manipulation and ICMP Time Exceeded messages.

In this case, the server was located within the same local network, so only one hop was detected.

Figure 11 – tracert command result

```
C:\>tracert empresa.com

Tracing route to 192.168.10.10 over a maximum of 30 hops:

  1    1 ms      0 ms      0 ms      192.168.10.10

Trace complete.
```

5. Analysis

During this lab, the proper operation of the following technologies and services was validated:

- DHCP
- DNS
- IP connectivity
- Name resolution
- Troubleshooting tools

It was observed that:

- ping verifies connectivity using ICMP

- nslookup verifies DNS resolution
- tracert identifies the route between devices

It was also demonstrated that a network can maintain functional IP connectivity even while the DNS service is failing.

Additionally, a possible corporate scenario was analyzed in which a poorly connected home router could begin assigning incorrect IP addresses through DHCP, generating connectivity issues and preventing access to internal resources.

6. Conclusion

This lab made it possible to understand the operation of essential network services such as DHCP and DNS, along with key diagnostic tools widely used in technical support and network administration.

The tests performed helped differentiate physical connectivity problems from issues related to name resolution.

In addition, the lab reinforced the use of troubleshooting commands commonly used in real-world environments, such as:

- ipconfig
- ping
- nslookup
- tracert

7. Simulation Considerations

It is important to note that Cisco Packet Tracer has certain limitations regarding the real-world behavior of some diagnostic tools.

In particular:

- Some advanced commands such as Test-NetConnection are not supported
- tracert behavior may be simplified in small topologies
- DNS simulation is more basic than in real environments

Despite these limitations, the tool allows a proper understanding of the fundamental concepts of troubleshooting and network services.